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FabAsset: Unique Digital Ass System for Hyperledg

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***Abstract*—Business is innovating with the advent of blockchain that tokenizes digital assets. To expand the blockchain's potential, Ethereum, a representative permissionless blockchain platform, supports the fungible token (FT) standard ERC-20 and the non-fungible token (NFT) standard ERC-721. Hyperledger Fabric (Fabric), a representative permissioned blockchain platform, proposed FabToken to support tokens in version 2.0.0 alpha. But FabToken contains only FTs, not NFTs. Given the market share in the enterprise blockchains, Fabric needs to support NFTs as soon as possible. This paper presents a unique digital asset management system called FabAsset so that Fabric can run decentralized applications that require NFTs. This paper describes the design of FabAsset, consisting of chaincode and SDK (Software Development Kit), and the prototype of a decentralized signature service leveraging FabAsset to validate its usefulness.**

***Index Terms*—permissioned blockchain; decentralized application; smart contract; non-fungible token;**

I. Introduction

Blockchain is a distributed ledger system that immutably records transactions maintained by nodes participating in a P2P network without a trusted authority [1]. Blockchain was used primarily as a technology for cryptocurrencies such as Bitcoin [2]. Ethereum [3], a representative permissionless blockchain platform, introduced smart contracts for the first time among blockchains. Smart contracts enable decentralized applications (dApps) to run on the blockchain and issue tokens for specific purposes.

Tokens represent digital assets governed by smart contracts. Tokens are classified into fungible tokens (FTs) and non-fungible tokens (NFTs) in terms of fungibility, which means whether one token can be replaced with another token of the same type and quantity. FTs are interchangeable with other FTs because they have the same value as others, and can be divided into smaller units. NFTs are not interchangeable with other NFTs because every NFT is a unique and indivisible unit itself. Ethereum adapts token standards such as ERC-20 [4] for FTs and ERC-721 [5] for NFTs to guarantee interoperability between dApps.

Tokens have revolutionized business. Initial coin offerings (ICOs) [6] introduced a new way of fundraising using ERC-20 tokens [7]. ICOs are referred to as crowd sales where startups trade their ERC-20 tokens with ethers. ICOs democratize the fundraising for startups, expanding the startup investment

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